CNI Plugin Setup on kubeadm

Calico, Flannel and Cilium (replacing Weave Net)

Important: Run only **one** CNI at a time. Before switching from one to another, always remove the old one and clean its leftovers (Section 0). The exception is Canal (Flannel networking plus Calico network policy), covered in Section 7.

0. Common cleanup (run before switching to ANY CNI)

Master node: remove the old CNI (run only the one that was installed)

```
# Weave
kubectl delete -f https://github.com/weaveworks/weave/releases/download/v2.8.1/weave-daemonset-k8s.yaml
# Calico
kubectl delete -f https://raw.githubusercontent.com/projectcalico/calico/v3.31.0/manifests/calico.yaml
# Flannel
kubectl delete -f https://github.com/flannel-io/flannel/releases/latest/download/kube-flannel.yml
# Cilium
cilium uninstall
```

Both master and worker: clean leftovers

```
sudo rm -f /etc/cni/net.d/*
sudo rm -rf /var/lib/weave /var/lib/cni /var/lib/calico /run/flannel /var/run/cilium

# delete leftover network interfaces
for i in weave cni0 flannel.1 vxlan.calico cilium_host cilium_net cilium_vxlan; do
  sudo ip link delete $i 2>/dev/null || true
done

# flush old iptables rules
sudo iptables -F && sudo iptables -t nat -F && sudo iptables -t mangle -F && sudo iptables -X

sudo modprobe br_netfilter
sudo modprobe overlay
sudo systemctl restart containerd
sudo systemctl restart kubelet
```

A reboot of the nodes is the cleanest option if the old CNI was Weave or Calico.

Then restart the pods that were using the old network

```
kubectl delete pod -n kube-system -l k8s-app=kube-dns
```

1. Pod CIDR (check this first)

Each CNI expects a pod network CIDR. Check what your cluster was created with:

```
kubectl cluster-info dump | grep -m1 cluster-cidr
kubectl get node -o jsonpath='{.items[*].spec.podCIDR}'
```

CNI	Default pod CIDR	Notes
Calico	192.168.0.0/16	Picks up the kubeadm CIDR automatically if set
Flannel	**10.244.0.0/16**	Must match `kubeadm init --pod-network-cidr=10.244.0.0/16`
Cilium	10.0.0.0/8 (cluster-pool)	Manages its own IPAM, no kubeadm CIDR needed

If Flannel is used and the cluster was created with a different CIDR, either edit the `net-conf.json` section of `kube-flannel.yml` to match, or re-init the cluster with `--pod-network-cidr=10.244.0.0/16`.

2. Calico (your existing setup)

```
# Master
kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.31.0/manifests/calico.yaml

kubectl get pods -n kube-system -w
kubectl get nodes
```

AWS security group rules (source: same security group)

- TCP 179 (BGP)
- IP protocol 4 (IP-in-IP), or UDP 4789 (VXLAN) if you use VXLAN mode
- Simplest option: allow **All traffic** from the same security group

3. Flannel

```
# Master
kubectl apply -f https://github.com/flannel-io/flannel/releases/latest/download/kube-flannel.yml

kubectl get pods -n kube-flannel -w
kubectl get nodes
```

Note that Flannel pods run in the `kube-flannel` namespace, not `kube-system`.

Prerequisites on every node

```
sudo modprobe br_netfilter
echo "br_netfilter" | sudo tee /etc/modules-load.d/k8s.conf
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables  = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward                 = 1
EOF
sudo sysctl --system
```

AWS security group rules (source: same security group)

- UDP 8472 (VXLAN, the default Flannel backend)
- or just allow **All traffic** from the same security group

Troubleshooting

```
kubectl get pods -n kube-flannel -o wide
kubectl logs -n kube-flannel -l app=flannel --tail=30
```

A common error is *Failed to check br_netfilter* or *node has no podCIDR*. The first means the module is missing; the second means the CIDR flag was not set at `kubeadm init`.

4. Cilium

Install the Cilium CLI (master node)

```
CILIUM_CLI_VERSION=$(curl -s https://raw.githubusercontent.com/cilium/cilium-cli/main/stable.txt)
CLI_ARCH=amd64   # use arm64 for Graviton instances
curl -L --fail --remote-name-all \
```

```
  https://github.com/cilium/cilium-cli/releases/download/${CILIUM_CLI_VERSION}/cilium-linux-${CLI_ARCH}.tar.gz{,.sha256sum
sha256sum --check cilium-linux-${CLI_ARCH}.tar.gz.sha256sum
sudo tar xzvfC cilium-linux-${CLI_ARCH}.tar.gz /usr/local/bin
rm cilium-linux-${CLI_ARCH}.tar.gz{,.sha256sum}
```

Install Cilium

```
cilium install
# or pin a version:  cilium install --version <x.y.z>
```

Verify

```
cilium status --wait
kubectl get pods -n kube-system -l k8s-app=cilium -w
kubectl get nodes
cilium connectivity test    # optional end-to-end test
```

Alternative using Helm

```
helm repo add cilium https://helm.cilium.io/
helm repo update
helm install cilium cilium/cilium -n kube-system
```

Requirements

- Linux kernel 5.4 or newer (check with `uname -r`)
- If you want Cilium to replace kube-proxy, the cluster must be created with `kubeadm init --skip-phases=addon/kube-proxy`. Otherwise leave the default.

AWS security group rules (source: same security group)

- UDP 8472 (VXLAN)
- TCP 4240 (health checks), ICMP
- TCP 4244 (Hubble, optional)
- or simply allow **All traffic** from the same security group

5. Common verification (for any CNI)

```
kubectl get nodes                      # all nodes should be Ready
kubectl get pods -A -o wide            # coredns should be Running with pod IPs
kubectl run test --image=busybox --restart=Never -- sleep 3600
kubectl exec test -- ping -c 3 <pod-ip-on-other-node>
kubectl delete pod test
```

6. If it doesn't work

> Check that the AWS security group allows **All traffic from the same security group**. Without this, pod-to-pod traffic across nodes fails for every CNI.

Then run these and share the output:

```
kubectl get pods -A -o wide

# Calico
kubectl describe pod -n kube-system -l k8s-app=calico-node | tail -30
# Flannel
kubectl describe pod -n kube-flannel -l app=flannel | tail -30
# Cilium
kubectl describe pod -n kube-system -l k8s-app=cilium | tail -30
```

```
journalctl -u kubelet --no-pager | tail -30
```

7. Optional: Flannel + Calico together (Canal)

Flannel and Calico cannot both be installed as full CNIs. The supported combination is **Canal**, where Flannel handles networking and Calico enforces network policies. It needs the `10.244.0.0/16` pod CIDR:

```
kubectl apply -f https://raw.githubusercontent.com/projectcalico/calico/v3.31.0/manifests/canal.yaml
```

Quick comparison

	Calico	Flannel	Cilium
Network policy	Yes	No	Yes (advanced, L7)
Complexity	Medium	Very low	Medium to high
Dataplane	iptables / BGP / VXLAN	VXLAN	eBPF
Best for	General production	Simple labs	Observability, advanced security